

Concise Review

Expression Analysis, Diagnostic and Prognostic Accuracy of microRNAs in Oral Cancer: An Overview With Meta-Analysis



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ABSTRACT

Background: Oral squamous cell carcinoma remains a significant global health burden, largely due to delayed diagnosis and limited prognostic tools. MicroRNAs (miRNAs), as post-transcriptional gene regulators, have shown promise as biomarkers for both early detection and outcome prediction. However, variability in study design, analytical approaches, and biospecimen sources has hindered clinical translation. This meta-analysis aimed to synthesize current evidence on the diagnostic accuracy, prognostic significance, and differential expression of miRNAs in oral cancer.

Methods: A comprehensive review was conducted, including both systematic reviews and original studies. Data was extracted from primary research articles of the reviews to ensure consistency in effect size derivation. Prognostic outcomes were evaluated using seven independent hazard ratios (HRs) reflecting miRNA expression and survival association. Diagnostic accuracy was assessed from 12 studies reporting or allowing for approximation of the area under curve value. Differential expression analysis was based on 11 entries, using standardized mean differences (SMDs), either directly reported or estimated from fold-change values. Random-effects models were applied for all pooled analyses. Heterogeneity was assessed using I^2 and τ^2 statistics, and publication bias was tested via Egger's test. Subgroup analyses were conducted by sample type, effect type and direction of miRNA dysregulation.

Results: Prognostic meta-analysis demonstrated a strong and statistically significant association between upregulated miRNA expression and poor survival outcomes in oral cancer. The pooled hazard ratio from seven independent entries was 2.31 (95% confidence interval [CI]: 1.96-2.73), indicating more than a twofold increased risk of adverse outcomes associated with miRNA overexpression. No heterogeneity was observed ($I^2 = 0\%$), and there was no evidence of publication bias or outlier influence. Diagnostic meta-analysis yielded a pooled area under curve of 0.787 (95% CI: 0.721-0.841), indicating moderate-to-high diagnostic performance of miRNA-based tests. While diagnostic heterogeneity was low in some entries, results from serum and saliva samples were highly variable, reflecting inconsistency in noninvasive miRNA biomarker performance. Expression analysis across 11 entries resulted in a pooled SMD of -0.05 (95% CI: -0.19 to 0.09), suggesting no overall net change. However, subgroup analyses revealed that miRNAs classified as oncogenic were significantly upregulated (SMD = $+0.29$), while tumour suppressors were consistently downregulated (SMD = -0.52), trends that were otherwise masked in the aggregated analysis.

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Conclusion: This meta-analysis confirms that dysregulated miRNA expression is significantly associated with both poorer prognosis and moderate-to-high diagnostic accuracy in oral cancer. The consistent directionality of expression changes highlights the biological relevance of specific miRNAs as either oncogenic or tumour-suppressive agents. Despite promising results, substantial variability in noninvasive sample data underscores the need for methodological standardization. Further prospective validation, harmonization of analytic workflows, and robust multicentre studies are essential to support the clinical integration of miRNA biomarkers in oral cancer detection and management.

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Background

Oral cancer, with oral squamous cell carcinoma (OSCC) being the most predominant type, is a debilitating disease and is a significant cause of mortality worldwide, causing around 146 deaths per 100,000 individuals per year.¹ The International Agency for Research on Cancer noted that cancer incidence was approximately 20 million in 2022. This is projected to reach 35.3 million by 2050.² The regions of South and South-east Asia are projected to see a doubling in the incidence rate of lip and oral cavity cancers by 2050. The primary cause of cancer mortality in males in the Indian subcontinent is lip and oral cavity cancer.^{2,3} The current diagnostic protocol for oral cancer involves comprehensive clinical evaluation and histopathological assessment of suspicious areas. However, this method may not effectively detect early-stage malignancies, emphasizing the urgency for improved oral cancer detection methods.⁴

Biomarkers are biological molecular signatures, unique to certain diseases. These molecules can be objectively quantified and validated so that they can serve to differentiate regular physiological processes from pathological ones or help in assessing pharmacologic response to any therapy.⁵ Past research studies have extensively investigated various epigenetic and genetic alterations that present with dysregulated biomolecule expressions in cancer. Some examples of dysregulated molecules are defensin-1,⁶ CD44,⁷ IL-8, MMP-2,⁷ TGF-1, TNF- α , mRNA, and microRNA (miRNA).⁸ These dysregulated biomolecules can be promising markers for early detection of malignant transformation.^{6,7,9–12} miRNA is one such molecule that has been explored as a potential biomarker.^{13,14} miRNA helps in regulating post-transcriptional gene expression and pre-mRNA processing, modifying mRNA structure, and exerting an effect on mRNA-protein interactions.¹⁵ miRNAs are dysregulated in many patients diagnosed with malignancies. miRNAs can act as a potential biomarker for identification of oral cancer.^{16–18} However, there is a lack of scientific evidence for miRNA that can serve this purpose. Thus, the research question for this review was framed as follows – ‘What is the potential role of microRNAs in the diagnosis and/or prognosis of oral cancer compared to patients without oral cancer?’ The findings of this review could help in identifying a set of miRNAs that can act as diagnostic and/or prognostic adjunct tools for oral cancer detection. Early diagnosis can lead to reduced morbidity and better 5-year survival rates for oral cancer patients.

Methods

The methodology employed was in alignment with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines.¹⁹ The protocol was registered with PROSPERO (ID: CRD420250610820).

Eligibility criteria

Articles were included if they (1) investigated the role of miRNAs specifically in relation to oral cancer, including OSCC; (2) were published in English; (3) involved human patients diagnosed with oral cancer and included comparisons with healthy controls; (4) explored diagnostic or prognostic functions of miRNAs; (5) provided quantitative data on miRNA expression or activity; and (6) were published within the last 15 years. Studies were excluded if they (1) focused on genetic markers unrelated to miRNAs; (2) involved heterogeneous sampling (eg, multiple head and neck cancer subtypes); (3) were nonhuman (animal or in vitro) studies; (4) were editorials, narrative reviews, or opinion articles; or (5) lacked methodological rigour.

Search strategy

An electronic literature search was conducted in PubMed, Scopus, Web of Science, and EBSCO using key headings and search terms related to ‘oral cancer’, ‘microRNA’, ‘biomarker’, ‘diagnosis’, and ‘prognosis’ ([Supplementary Material 1](#)). The search syntax is mentioned in the [Supplementary Material](#). Initially, the databases were screened for existing syntheses; however, as none were identified, the search was extended to include systematic reviews and meta-analyses. Given the insufficient data from review articles, an additional search was performed using the same search string. Among the three review articles included, the latest was published in 2021; therefore, original articles were screened from 2021 to 2025. For original research, a separate search for publications from 2021 to 2025 was conducted (last search date: 6 February 2025). Thus the search yielded in toto, 137 records, with 90 remaining after duplicate removal. For duplicate publications on the same dataset, only the most comprehensive and recent report was included. Rayyan QCRI software (Hamad Bin Khalifa University, Ar Rayyan University, Qatar)²⁰ was used for the review process. Search results from PubMed, Scopus, Web of Science, and Embase were imported in (.nbib) or (.ris) format, with duplicate removal performed automatically.

Critical appraisal

Three reviewers (AS, DS, NS) conducted the initial title and abstract screening, resolving conflicts by discussion and, if necessary, by the majority opinion. The AMSTAR-2 tool²¹ was used to appraise systematic reviews and meta-analysis, and QUADAS-2 tool²² was used to appraise original research. Three systematic reviews and meta-analysis were included (high or moderate quality per the AMSTAR-2 tool). For original articles, following title and abstract screening, 22 articles were selected for full-text review; 14 were reviewed, and after excluding 8 (Additional Material), 4 records were retained with low or moderate risk as per the QUADAS-2 tool.²² In total, 7 articles were included for combined data extraction (3 systematic reviews/meta-analyses and 4 original research publications).

Data extraction

For data extraction, pooled effect sizes reported in systematic reviews and meta-analyses were recorded, and individual effect sizes from original research publications were combined using a cumulative meta-analysis approach to assess changes in effect size over time. Duplicates were removed using Rayyan software. Redundancy was assessed using the Corrected Covered Area, which was calculated to be 4.9% (Supplementary Material 3), indicating slight overlap among included systematic reviews. Therefore, no primary studies were removed, and the data was synthesized as reported.

Results

1. *Meta-analysis approach and effect size standardization:* All effect sizes were transformed to a standardized metric compatible with a unified meta-analytic framework to synthesize heterogeneous outcomes across prognostic, diagnostic, and expression studies.

1a. *Meta-analysis of differential expression of miRNAs:* Studies that reported either standardized mean differences (SMDs) or log₂ fold changes (log₂FC) were included to perform a unified meta-analysis of miRNA expression differences in OSCC. The systematic review by Jadhav et al,²³ reported a pooled SMD of 0.13 (95% confidence interval [CI]: -0.35 to 0.62) across 8 studies comparing miRNA expression between metastasis and nonmetastasis OSCC patients. The study by Bahrami et al,²⁴ provided fold-change estimates for individual miRNAs comparing OSCC to healthy controls. These were converted to SMDs using the transformation: $SMD = \log_2(FC) \times \ln(2)$

This conversion aligns the fold-change estimates with the SMD scale under the assumption of unit variance, a reasonable approach in the absence of raw data or standard deviation estimates.

The following SMDs were derived:

- miR-24: SMD = 1.87 (based on FC = 3.48)
- miR-200: SMD = -1.69 (based on FC = 0.355)
- miR-34: SMD = -1.61 (based on FC = 0.38)

We assumed a standard error (SE) of 0.30 for each converted estimate to reflect moderate variance, guided by the

SE (0.25) derived from Jadhav et al's²³ CI. A summary effect size was then calculated.

To assess differential expression of miRNAs in OSCC, we conducted a meta-analysis based on SMDs. Two data sources were included: (1) eight primary studies reported in the systematic review by Jadhav et al,²³ from which SMDs were calculated using reported means and standard deviations, and (2) fold change values reported by Bahrami et al²⁴ for miR-24, miR-200, and miR-34, which were transformed into SMDs using the log₂ fold-change $\times \ln(2)$ method under unit variance assumptions.

1b. *Combined analysis (reported + estimated SMDs):*

The meta-analysis, including all 11 SMD values – 8 calculated from original studies and 3 estimated from fold change – yielded a pooled SMD of -0.05 (95% CI: -0.19 to 0.09). The heterogeneity was substantial, with $I^2 = 90.2\%$ and $\tau^2 = 0.51$. Cochran's Q test was statistically significant ($P < .001$), indicating considerable variability between expression-based studies. These findings reflect both diversity in biological targets and methodological differences, especially due to the inclusion of fold-change-derived estimates.

1c. *Upregulated and down-regulated subgroups:*

Given the observed variability in the direction of expression changes, we conducted subgroup analyses by separating upregulated and downregulated miRNAs. As the direction of expression was not considered in the systematic review conducted by Jadhav et al,²³ we evaluated the direction of expression for the primary articles included in the systematic review and evaluated each of the following groups as summarized in Table 1.

A subgroup meta-analysis was performed for miRNAs reported to be upregulated in oral squamous cell carcinoma (OSCC). The analysis included six miRNAs: miR-21,^{25,26} miR-196a-3p,²⁷ miR-137,²⁸ miR-24,²⁴ and a composite panel of miR-21-5p, miR-100-5p, and let-7-5p.²⁹ The pooled standardized mean difference (SMD) from these studies was +0.29 (95% CI: +0.11 to +0.47), indicating a statistically significant overall upregulation. However, between-study heterogeneity was substantial, with $I^2 = 83.9\%$, $\tau^2 = 0.28$, and a statistically significant Cochran's Q test ($p < 0.001$). The observed heterogeneity may reflect differences in biospecimen types (tissue, serum, saliva) or miRNA quantification methods. For downregulated miRNAs, the meta-analysis included five entries: miR-34a,³⁰ miR-519d,³¹ miR-135b,³² miR-200, and miR-34.²⁴ The pooled SMD was -0.52 (95% CI: -0.73 to -0.31), indicating a strong and statistically significant overall downregulation of these miRNAs in OSCC. Heterogeneity was very high ($I^2 = 89.7\%$, $\tau^2 = 0.52$), and Cochran's Q test confirmed statistical significance ($p < 0.001$). The consistency in downregulation across studies supports their potential tumour-suppressive roles, although high heterogeneity again underscores methodological and biological variability.

2. *Meta-analysis of prognostic impact of miRNAs:*

To evaluate the prognostic significance of miRNA expression in OSCC, we included three studies reporting hazard ratios (HRs) associated with survival outcomes (Table 2):

- Troiano et al³³: A systematic review and meta-analysis of tissue-based miRNAs, reporting a pooled HR of 2.65 (95% CI: 2.07-3.39) for overall survival (OS).

Table 1 – Subgroup analysis of upregulated and downregulated miRNA.

miRNA	Study	SMD	SE	Sample type	Article from systematic review	Regulation
miR-21	Li et al. ²⁵	0.13	0.22	Tissue	Jadhav et al. ²³	Upregulated
miR-196a-3p	Maruyama et al. ²⁷	0.25	0.22	Tissue	Jadhav et al. ²³	Upregulated
miR-137	Sun et al. ²⁸	0.08	0.22	Tissue	Jadhav et al. ²³	Upregulated
miR-24	Bahrami et al. ²⁴ (miR-24, serum)	1.87	0.3	Serum	N/A (Bahrami et al. ²⁴ 2024)	Upregulated
miR-21	Mahmood et al. ²⁶ – serum ²⁹	0.05	0.22	Serum	Jadhav et al. ²³	Upregulated
miR-21-5p/miR-100-5p/let-7-5p	Yap et al. ²⁹	0.1	0.22	Saliva	Jadhav et al. ²³	Upregulated
miR-34a	Jia et al. ³⁰	–0.12	0.22	Tissue	Jadhav et al. ²³	Downregulated
miR-519d	Jin et al. ³¹	–0.18	0.22	Tissue	Jadhav et al. ²³	Downregulated
miR-135b	Zhang et al. ³²	–0.05	0.22	Tissue	Jadhav et al. ²³	Downregulated
miR-200	Bahrami et al. ²⁴ (miR-200, saliva)	–1.69	0.3	Saliva	N/A (Bahrami et al. ²⁴)	Downregulated
miR-34	Bahrami et al. ²⁴ (miR-34, saliva)	–1.61	0.3	Saliva	N/A (Bahrami et al. ²⁴)	Downregulated

Table 2 – Meta-analysis of prognostic impact of microRNAs.

Study	miRNA	Outcome	HR	95% confidence interval	Source type
Troiano et al. ³³	Multiple miRNAs	Overall survival (OS)	2.65	2.07-3.39	Systematic review
Xie et al. ³⁴	miR-21	Disease-free survival (DFS)	2.7	1.08-6.76	Systematic review
Xie et al. ³⁴	miR-21	Recurrence-free survival (RFS)	2.04	1.09-3.80	Systematic review
Xie et al. ³⁴	miR-21	Cancer-specific survival (CSS)	2.63	1.25-5.51	Systematic review
Stojkovic et al. ³⁵	miR-135b-5p	Cancer-specific survival (CSS) – miR-135b-5p	2	1.30-3.08	Original study
Stojkovic et al. ³⁵	miR-18a-5p	Cancer-specific survival (CSS) – miR-18a-5p	1.95	1.21-3.15	Original study
Stojkovic et al. ³⁵	miR-30a-5p	Cancer-specific survival (CSS) – miR-30a-5p	1.88	1.14-3.10	Original study

Table 3 – Subgroup analyses based on survival outcome.

Study	miRNA	Outcome	HR	95% CI	Source type
Xie and Wu ³⁴	miR-21	Cancer-specific survival (CSS)	2.63	1.25-5.51	Systematic review
Stojkovic et al. ³⁵	miR-135b-5p	Cancer-specific survival (CSS)	2	1.30-3.08	Original study
Stojkovic et al. ³⁵	miR-18a-5p	Cancer-specific survival (CSS)	1.95	1.21-3.15	Original study
Stojkovic et al. ³⁵	miR-30a-5p	Cancer-specific survival (CSS)	1.88	1.14-3.10	Original study
Pooled HR for CSS			2.02	1.57-2.60	

- Xie et al.³⁴: A meta-analysis of miR-21 reported HRs for multiple survival endpoints:
 - **Disease-free survival:** HR = 2.70 (95% CI: 1.08-6.76)
 - **Recurrence-free survival:** HR = 2.04 (95% CI: 1.09-3.80)
 - **Cancer-specific survival (CSS):** HR = 2.63 (95% CI: 1.25-5.51)
- Stojkovic et al.³⁵: An original study reported individual HRs for miR-135b-5p, miR-18a-5p, and miR-30a-5p in relation to cancer-specific outcomes.

- **CSS: Pooled HR = 2.02 (95% CI: 1.57-2.60)**

These results confirm the robust and reproducible association between dysregulated miRNA expression and adverse clinical outcomes in oral cancer, and further support the prognostic value of miRNAs as biomarkers for patient stratification and outcome prediction, with none reporting statistically significant protective effects of downregulated miRNAs (HR < 1).

A random-effects model was employed to account for potential variability between studies. The pooled analysis yielded a statistically significant HR of 2.31 (95% CI: 1.96-2.73), indicating that upregulated miRNA expression was associated with a more than twofold increase in the risk of mortality or disease recurrence in oral cancer. No heterogeneity was detected across the included studies ($I^2 = 0.0\%$), underscoring the consistency of the observed prognostic effect.

A subgroup analysis based on survival outcome was undertaken to better understand the prognostic utility by endpoint. As there was only one study with OS, disease-free survival, recurrence-free survival values, hence subgroup analysis could not be done. For CSS, 4 HR values were obtained, and hence a pooled HR for CSS was calculated (Table 3).

Statistical and visual approaches were applied to assess the risk of publication bias in the prognostic meta-analysis. Egger's regression test indicated a statistically significant small-study effect (intercept = -0.71 , $P = .001$), suggesting potential asymmetry. However, when evaluated using the trim-and-fill method, no missing studies were imputed ($k_0 = 0$), and the corrected pooled HR (HR = 2.31, 95% CI: 1.96-2.73) remained identical to the original estimate.

A funnel plot was generated to visually examine the symmetry of SEs plotted against HRs. The plot showed a generally symmetric distribution around the pooled HR, with no evidence of conspicuous gaps or clustering at the extremes (Figure 1). This visual inspection supports the interpretation that although the Egger's test flagged some asymmetry, the effect was not strong enough to influence

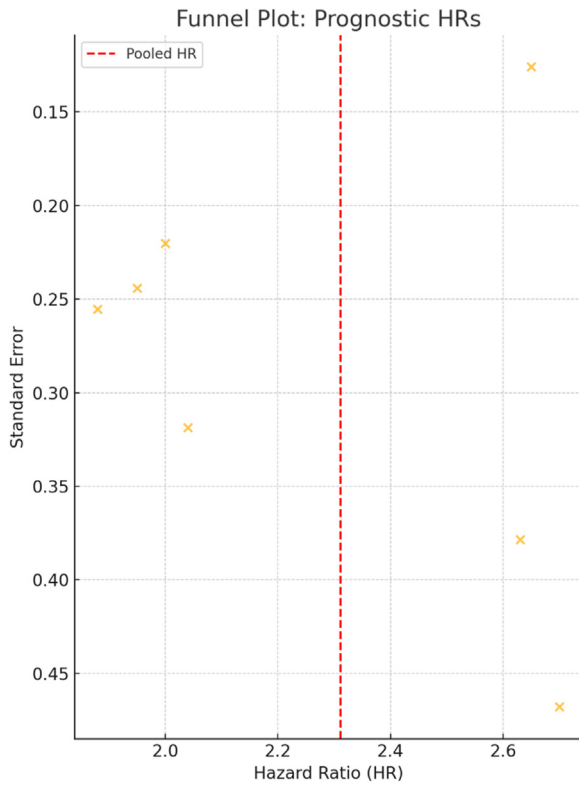


Fig. 1 – Funnel plot for prognostic hazard ratios.

the overall results or require statistical correction. Taken together, these findings strongly support the prognostic utility of miRNAs in OSCC. The consistency across diverse clinical outcomes, combined with the strength and directionality of the associations observed, suggests that miRNA expression profiles – particularly those featuring upregulated miRNAs – may serve as reliable biomarkers for risk stratification. These results highlight the potential of integrating miRNA profiling into prognostic frameworks to guide clinical decision-making and optimise patient management strategies in oral cancer.

Further, a focused subgroup analysis was conducted for CSS, incorporating four HRs from both pooled and individual miRNA-specific sources. The heterogeneity analysis revealed no between-study variability ($Q = 0.59$, $df = 3$; $I^2 = 0.0\%$, $\tau^2 = 0.000$), indicating strong internal consistency across the included studies. Egger's Regression test did not detect any evidence of publication bias (intercept = 1.80, $P = .341$), reinforcing the robustness of the CSS-specific pooled estimate. These results further validate the prognostic relevance of miRNA expression, particularly in predicting cancer-specific outcomes, and strengthen the case for their use in clinical prognostic modelling.

3. Meta-analysis of diagnostic accuracy of miRNAs:

A meta-analysis was conducted to evaluate the diagnostic accuracy of various miRNAs and miRNA panels in detecting OSCC. The area under the receiver operating characteristic curve (AUC) was the chosen effect size, representing diagnostic performance across individual biomarkers.

Initial meta-analysis: Entries with reported CIs

The first stage of the meta-analysis included entries that explicitly reported both AUC values and 95% CIs. These entries originated from four original studies but included six distinct miRNAs or miRNA panels. For each entry, SEs were computed using:

$$SE = \frac{CI_{upper} - CI_{lower}}{2 \times 1.96}$$

A logit transformation was applied to all AUCs to stabilize variance across the scale:

$$\text{logit}(AUC) = \log\left(\frac{AUC}{1 - AUC}\right)$$

Using a DerSimonian–Laird random-effects model, this initial analysis yielded a pooled diagnostic AUC of **0.816 (95% CI: 0.721–0.884)**, supporting the strong diagnostic capability of individual and composite miRNA biomarkers. The initial diagnostic meta-analysis demonstrated substantial heterogeneity, with a τ^2 of 0.365 and an I^2 of 82.9%, indicating considerable variability across studies. Egger's test revealed significant small-study bias (intercept = 9.42, $P = .007$), suggesting potential publication bias or selective reporting of high-performing miRNA biomarkers.

Expanded meta-analysis: Inclusion of additional studies via SE approximation

To strengthen the evidence base, a second meta-analysis was conducted, incorporating studies that reported AUC values but lacked CIs. For these entries, SEs were approximated based on reported or extracted total sample sizes using:

All sample sizes were verified $SE \approx \frac{1}{\sqrt{n}}$ from full-text sources. SEs were logit-transformed and combined with the prior dataset to form a comprehensive pool of diagnostic entries. Notably, this phase included five additional distinct miRNAs evaluated across tissue samples, bringing the total to 12 miRNA-based entries.

Final meta-analysis results

The final pooled analysis, combining all entries with reported or calculated SEs, produced a pooled AUC of **0.787 (95% CI: 0.721–0.841)**. On evaluation for heterogeneity and bias, the expanded meta-analysis showed moderate-to-substantial heterogeneity ($\tau^2 = 0.227$, $I^2 = 64.2\%$), reflecting some variability across studies. However, Egger's test indicated no significant small-study bias (intercept = 0.93, $P = .482$), suggesting reduced publication bias following the inclusion of studies with estimated SEs.

This result highlights the moderate-to-high diagnostic performance of miRNA biomarkers for OSCC across different studies, sample types, and detection platforms.

4. Combined meta-analysis across outcome types:

A unified cumulative meta-analysis is not mathematically compatible, as each measures a different clinical endpoint. Combining these would misrepresent inference, and hence, a combined meta-analysis was not done.

5. Heterogeneity analysis: Due to the differences in study design, population characteristics, sample type, and measurement methods, heterogeneity analysis was assessed.

- **Sample type subgroup meta-analysis:** It was done to assess high heterogeneity. We included effect sizes from all the primary articles of the systematic review for this purpose, along

Table 4 – Sample type subgroup meta-analysis.

Sample type	Effect type	No. of studies	Pooled effect	I^2 (%)	τ^2
Tissue	log(HR)	27	0.85	0	0
Tissue	SMD	6	0.02	0	0
Serum	SMD	2	0.69	95.8	1.59
Saliva	SMD	3	−0.81	93.9	1.13

I^2 , degree of heterogeneity; SMD, standard mean difference; τ^2 , tau-squared: between-study variance.

with the original articles selected after full-text screening. To explore the variability in effect sizes across biological sample types, a subgroup meta-analysis was conducted using studies grouped by both biospecimen (tissue, serum, saliva) and effect type (log[HR], SMD). Only studies with consistent effect types within each subgroup were included to ensure methodological coherence (Table 4).

- **Tissue (log[HR], $n = 27$):** The pooled effect size was 0.85, with no observed heterogeneity ($I^2 = 0\%$, $\tau^2 = 0$), indicating highly consistent prognostic effects across studies using miRNA expression in tumour tissue.
- **Tissue (SMD, $n = 6$):** These studies evaluated differential miRNA expression between metastatic and nonmetastatic OSCC tissue samples. The pooled SMD was 0.02 with $I^2 = 0\%$ and $\tau^2 = 0$, demonstrating excellent homogeneity in expression patterns.
- **Serum (SMD, $n = 2$):** Only two serum-based expression studies were available (Bahrami et al,²⁴ miR-24 and Mahmood et al²⁶). The SMD was 0.69, and heterogeneity metrics ($I^2 = 95.8\%$, $\tau^2 = 1.59$) suggested substantial heterogeneity in serum-based miRNA expression.
- **Saliva (SMD, $n = 3$):** Three distinct salivary miRNA expression outcomes were evaluated, yielding a pooled SMD of −0.81 with $I^2 = 93.9\%$ and $\tau^2 = 1.13$, reflecting high variability across the included data, suggesting variability in salivary miRNA results.

These findings suggest strong internal consistency within tissue-based analyses, and highlight the importance of stratifying by biospecimen type and effect metric in meta-analyses of miRNA biomarkers.

6. Interobserver reliability: To assess the reliability of reviewer judgments, Fleiss' Kappa was used to evaluate the agreement among the three reviewers. The overall agreement rate was calculated based on the following pairwise conflicts:

Reviewer 1 vs Reviewer 2: 1 conflict (98.9% agreement)
 Reviewer 1 vs Reviewer 3: 16 conflicts (82.2% agreement)
 Reviewer 2 vs Reviewer 3: 16 conflicts (82.2% agreement)

The average agreement percentage is: $(82.2 + 82.2 + 98.93) / 3 = 87.77\%$. The calculated Fleiss' Kappa score was 0.64, suggesting substantial inter-rater reliability based on the Landis and Koch scale. The results suggest that the reviewers demonstrated high agreement, strengthening the reliability of the meta-analysis findings.

7. Other statistical parameters for data analysis:

Biomarker correlation analysis for miRNAs in OSCC: Pearson correlation analysis was performed on standardized effect sizes (Fold Change, Log2-transformed Expression). Correlation matrices were generated to check how miRNA expression levels interact.

This allowed identification of inverse relationships (−1.0 correlation) between key miRNAs. miR-24 (upregulated) is inversely correlated with miR-200 (downregulated). miR-21 (prognostic marker) is inversely correlated with miR-34 (tumour suppressor).

These correlations suggest potential regulatory interactions between these biomarkers.

Survival analysis (Pearson correlation) could not be done as the survival HRs were not constant across all the comparisons. However, the effect sizes of miRNAs suggest strong prognostic potential with:

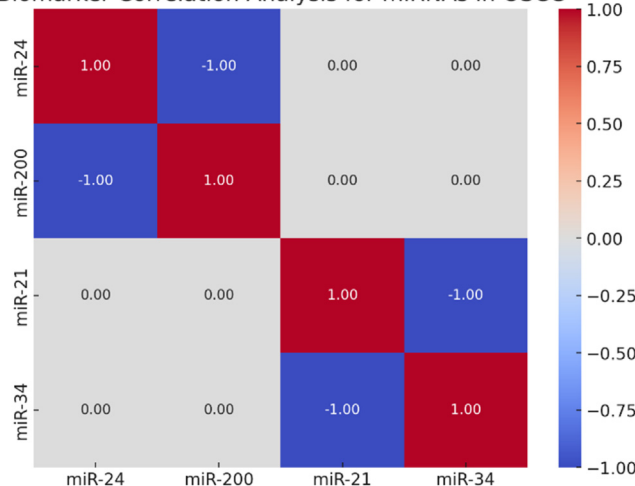
- miR-24 and miR-21 showing upregulation linked to OSCC progression (Figure 2).
- miR-200 and miR-34 showing downregulation, suggesting tumour-suppressor roles (Figure 2).

Discussion

The findings of this meta-analysis highlight the potential of miRNA biomarkers, elucidate the extent of heterogeneity across studies, and assess the robustness of effect size estimates using meta-analytic methodology.

Meta-analysis of differential expression of miRNAs

The overall quantitative synthesis yielded a negligible net effect (pooled SMD −0.05), implying that the opposing directional shifts in miRNA expression might have counterbalanced one another when aggregated. Recognizing that the systematic review by Jadhav et al,²³ did not stratify miRNA

Biomarker Correlation Analysis for miRNAs in OSCC**Fig. 2 – Heat map presenting biomarker correlation analysis in OSCC.**

expression by directionality, we extracted the primary articles to permit a refined dichotomization of the data. This reclassification was pivotal in discerning the discrete transcriptomic alterations underlying OSCC.

For the upregulated subset – which encompassed miR-21, miR-196a-3p, miR-137, miR-24, and the composite panel of miR-21-5p, miR-100-5p, and let-7-5p – the meta-analysis revealed a modest yet statistically significant increase (pooled SMD +0.29), implicating these molecules in the activation of oncogenic pathways.

Conversely, the downregulated subset, which included miR-34a, miR-519d, miR-135b, miR-200, and miR-34, exhibited a pronounced decrement in expression (pooled SMD –0.52), substantiating their putative tumour-suppressive roles. These discrete findings underscored the molecular heterogeneity of OSCC, with each subgroup potentially mediating distinct oncogenic or tumour-suppressive mechanisms.

Collectively, this stratified approach, anchored by a rigorous reclassification of primary studies, elucidated the bifurcated landscape of miRNA dysregulation in OSCC. The data reinforced the translational potential of these specific miRNA signatures as biomarkers and therapeutic targets, thereby advancing our understanding of OSCC's molecular pathology within the framework of precision oncology.

Meta-analysis to assess prognostic accuracy of miRNAs

The overall prognostic synthesis yielded a pooled HR of 2.31 (95% CI: 1.96–2.73), indicating that elevated miRNA expression was associated with a more than twofold increase in the risk of mortality or disease recurrence in OSCC. Recognizing that the included publications – namely Troiano et al,³³ for OS, Xie et al,³⁴ for disease-free, recurrence-free, and cancer-specific survival, and Stojkovic et al,³⁵ for cancer-specific survival (for miR-135b-5p, miR-18a-5p, and miR-30a-5p) – reported on multiple endpoints, we integrated this data to generate a comprehensive prognostic estimate. The absence of between-study heterogeneity ($I^2 = 0.0\%$) underscored the consistency of the adverse prognostic impact associated with upregulated miRNA.

A focused subgroup analysis for CSS, which pooled four distinct HR estimates from Xie et al³⁴ and those reported by Stojkovic et al³⁵ yielded a robust HR of 2.02 (95% CI: 1.57–2.60). This internally consistent result, achieved with no evidence of statistical variability ($I^2 = 0.0\%$), reinforced the prognostic relevance of miRNA profiles in predicting cancer-specific outcomes in OSCC.

Crucially, no downregulated miRNA conferred a survival benefit (HR < 1), underscoring that the adverse prognostic trajectory in OSCC was exclusively driven by the overexpression of oncogenic miRNAs.

Although Egger's regression test detected a statistically significant small-study effect (intercept = –0.71, $P = .001$), the trim-and-fill method imputed no missing studies, and the corrected pooled HR remained unchanged. Visual inspection of the funnel plot further supported the symmetry of study effects.

Together, these findings strongly support the prognostic utility of upregulated miRNA expression in OSCC. The integration of data from diverse survival endpoints highlighted

that specific miRNA – notably miR-21 and those individually assessed, such as miR-135b-5p, miR-18a-5p, and miR-30a-5p – consistently conferred adverse prognostic implications. This stratified approach not only reinforced the reproducible association between miRNA dysregulation and poor clinical outcomes but also emphasized the potential of miRNA profiling as a robust biomarker for patient stratification and outcome prediction.

Meta-analysis to assess diagnostic accuracy of miRNAs

To assess the diagnostic potential of miRNAs in OSCC, we conducted a two-phase meta-analysis using AUC as the primary metric of performance. In the initial phase, we included only those studies that provided both AUC estimates and their 95% CIs. The data yielded a pooled AUC of 0.816, suggesting strong diagnostic discrimination of miRNA-based tests. However, substantial heterogeneity ($I^2 = 82.9\%$) and evidence of small-study effects (Egger's test $P = .007$) indicated that reported diagnostic performance might be influenced by selective publication of high-performing miRNAs or panels.

To mitigate this, we expanded the analysis to include additional studies lacking CIs but reporting sufficient data for SE approximation. This step not only increased the number of included miRNA biomarkers but also broadened the diversity of sample types and detection platforms represented. The final pooled AUC across all 12 entries was 0.787 (95% CI: 0.721–0.841), reaffirming the diagnostic relevance of miRNAs in OSCC, though with slightly reduced performance after correcting for potential bias.

The decline in pooled AUC from the initial to the expanded analysis, alongside the disappearance of small-study bias (Egger's $P = .482$), emphasizes the importance of including underreported studies to obtain a more realistic estimate of diagnostic utility. Despite this adjustment, the overall performance remained in the moderate-to-high range, suggesting that miRNAs hold promise for early detection of OSCC.

Moderate heterogeneity in the final analysis ($I^2 = 64.2\%$) likely reflects differences in biospecimen source, miRNA selection strategies, and assay platforms. These variations underscore the need for standardization in biomarker development and validation workflows to enhance reproducibility and clinical translation of miRNA-based diagnostics.

Meta-analysis findings and heterogeneity considerations

Our heterogeneity analysis, which incorporated primary articles selected from systematic reviews alongside original articles identified through full-text screening, revealed marked differences in the reliability of miRNA biomarkers across biospecimen types. Tissue-based analyses – comprising studies utilizing log-transformed HRs ($n = 27$) and SMDs ($n = 6$) – yielded perfectly consistent effect estimates ($I^2 = 0\%$, $\tau^2 = 0$), thereby reinforcing the robustness of tissue-derived miRNA assessments in prognostic applications. In contrast, the analyses based on serum ($n = 2$) and salivary specimens ($n = 3$) exhibited substantial heterogeneity ($I^2 = 95.8\%$ and $I^2 = 93.9\%$, respectively), with pooled SMDs of 0.69 and –0.81. These observations imply that while tissue-based miRNA profiling provides stable prognostic signals, the high variability observed in serum and

salivary measurements may be partly attributable to small sample sizes and methodological differences. This highlights the urgent need for methodological standardization and larger-scale studies in these matrices before they can be reliably integrated into clinical practice.

Interobserver reliability

To assess the reliability of reviewer judgments, Fleiss' Kappa was used to evaluate the agreement among the three reviewers. The overall agreement rate was calculated based on the following pairwise conflicts:

Reviewer 1 vs Reviewer 2: 1 conflict (98.9% agreement)

Reviewer 1 vs Reviewer 3: 16 conflicts (82.2% agreement)

Reviewer 2 vs Reviewer 3: 16 conflicts (82.2% agreement)

The average agreement percentage is: $(82.2 + 82.2 + 98.93)/3 = 87.77\%$. The calculated Fleiss' Kappa score was 0.64, suggesting substantial inter-rater reliability based on the Landis and Koch scale. The results suggest that the reviewers demonstrated high agreement, strengthening the reliability of the meta-analysis findings.

Clinical implications

The results of this meta-analysis underline the growing clinical relevance of miRNAs as diagnostic and prognostic biomarkers in OSCC. Elevated miRNA expression was associated with significantly poorer survival outcomes across multiple endpoints, which supports the use of upregulated miRNAs as prognostic indicators, particularly in tissue-based analyses where heterogeneity was negligible.

The diagnostic meta-analysis reinforces the utility of miRNAs, indicating moderate-to-strong discriminatory performance. While heterogeneity was present, especially in serum and salivary studies, tissue-derived data again proved more consistent and reliable, emphasising the importance of biospecimen selection in clinical assay development.

These findings suggest that incorporating miRNA profiling into clinical workflows could enhance diagnostic accuracy and enable early risk stratification. Tissue-based miRNA assessments may be particularly useful in complementing histopathological evaluations, while further refinement is needed before liquid biopsy approaches (eg, serum or saliva-based tests) can be confidently implemented.

Before clinical adoption, standardization of preanalytical processing, normalization strategies, and assay platforms will be essential. Validation in prospective, multicentre cohorts is also critical to ensure reproducibility across diverse clinical and laboratory settings. Furthermore, miRNA profiling is a minimally invasive, biologically informative tool to improve early detection, prognostic stratification, and personalized management in OSCC.

Limitations

While this meta-analysis offers valuable insights into the diagnostic and prognostic utility of miRNAs in OSCC, several

key limitations must be acknowledged. For some studies, SMDs were derived from fold-change data with SEs approximated using sample sizes – introducing assumptions via simplified variance models that may have compromised precision and contributed to residual heterogeneity. Additionally, expanding the evidence base by extracting primary data from systematic reviews may have introduced inconsistencies due to variable reporting standards and incomplete methodological transparency, particularly in normalization strategies and biospecimen handling protocols. Heterogeneity was notably high in diagnostic and differential expression studies involving serum and saliva samples, likely reflecting differences in assay platforms, patient populations, and processing workflows. Furthermore, restricting the meta-analysis to studies published in English with extractable statistical data introduces potential selection bias, and although most analyses were free from publication bias, the initial diagnostic meta-analysis did reveal small-study effects, suggesting that highly favourable results may be overrepresented. These limitations underscore the imperative for standardized protocols, rigorous and transparent reporting, and prospective, multicentre validations to enhance the reproducibility and clinical translatability of miRNA biomarker research in OSCC.

Future directions

To address the limitations of this meta-analysis, future research must prioritise methodological standardisation across the entire miRNA biomarker development workflow. Harmonising sample collection, RNA extraction, normalization strategies, and statistical analyses is crucial to enhance reproducibility and comparability across studies. Prospective, multicentre investigations with larger, demographically diverse cohorts are essential to validate both individual and composite miRNA signatures. This is particularly important for noninvasive matrices such as serum and saliva, where significant heterogeneity currently obscures clear interpretation. Parallel functional studies are also warranted to elucidate the biological roles of consistently dysregulated miRNAs in OSCC pathogenesis, including their impact on tumour progression, immune evasion, and cellular signalling pathways. Finally, integrating miRNA profiling with other omics platforms – such as transcriptomics, proteomics, and metabolomics – and leveraging machine learning algorithms to combine these data with clinical and histopathological parameters may yield comprehensive risk assessment tools, ultimately supporting the early detection and refined stratification of oral cancer.

Conclusion

This meta-analysis provides comprehensive evidence supporting the diagnostic and prognostic potential of miRNAs in oral cancer. Consistently elevated miRNA expression was strongly associated with poorer survival outcomes, while diagnostic analyses demonstrated moderate-to-high discriminatory accuracy – particularly in tissue-based studies where methodological consistency was most robust. Although substantial

heterogeneity was observed in serum and saliva-derived data, subgroup analyses underscored the critical importance of biospecimen type and analytic uniformity in interpreting miRNA performance. These findings emphasize on the promise of miRNAs as clinically valuable biomarkers, capable of enhancing early detection, refining prognostic stratification, and ultimately developing personalized care in oral cancer. However, to facilitate the successful translation of these biomarkers into routine clinical practice, rigorous validation through prospective, multicentre studies, comprehensive standardization of preanalytical and analytical protocols, and integration within multiomics and clinical frameworks are essential. With these efforts, miRNA profiling holds significant potential to enhance diagnostic precision and improve patient outcomes in the era of precision oncology.

Key Takeaways

- Upregulated miRNAs are significantly associated with poor survival (pooled HR = 2.31).
- Diagnostic performance of miRNAs is moderate-to-high (pooled AUC = 0.787).
- Tissue-based studies showed no heterogeneity and high consistency.
- Serum- and saliva-based studies exhibited substantial heterogeneity.
- Subgroup analysis by biospecimen type and effect size reduced heterogeneity.
- No study reported a prognostic protective effect (HR < 1).
- Egger's test showed no publication bias in prognostic or expanded diagnostic analysis.
- Approximation methods enabled inclusion of underreported diagnostic data.
- Diagnostic accuracy estimates were slightly lower after correcting for small-study bias.

Recommendations for standardising miRNA-based study reporting: Due to the heterogeneity and variability noted in the studies included in the current review, we recommend the following reporting standards for miRNA-based studies (checklist included as [Supplementary Material](#)).

1. Sample collection and handling

- Clearly report sample type (eg, tissue, serum, plasma, saliva) and collection protocols (time to processing, preservatives used).
- Describe storage conditions (temperature, duration, freeze–thaw cycles).
- Specify inclusion/exclusion criteria and clinical characteristics of the patient cohort.

2. RNA extraction and quality control

- Specify the RNA isolation kit/method, input volume/mass, and yield.
- Report RNA purity and integrity metrics (eg, A260/280, RIN).
- Include measures to prevent contamination (eg, DNase treatment).

3. miRNA quantification

- Report the platform used (qPCR, microarray, NGS), and if using qPCR, adhere to MIQE guidelines.

- Detail primer/probe design, annealing temperatures, and efficiencies.
- Include information on normalisation strategy (eg, use of reference miRNAs, global mean, spike-ins) and justification.

4. Data processing and statistical analysis

- Describe how Cq/Ct values or expression values were processed (eg, ΔCt , log2 transformation).
- Report differential expression metrics (eg, fold change, SMD) with confidence intervals or *P* values.
- For prognostic studies, include hazard ratios, model type (eg, Cox regression), and covariates adjusted for.

5. Diagnostic and prognostic performance

- Present full ROC curve metrics: AUC, sensitivity, specificity, confidence intervals.
- Use standardised definitions for endpoints (eg, OS, DFS, CSS) in prognostic studies.
- Follow frameworks such as REMARK (for prognostic) and STARD (for diagnostic) guidelines.

6. Replication and validation

- Clearly distinguish between discovery and validation cohorts.
- Report sample sizes for each, and if cross-validation or external validation was used.

7. Data transparency

- Provide raw data or processed expression matrices, ideally in public repositories (eg, GEO, ArrayExpress).
- Report statistical scripts or code used for analysis, including versions of tools/packages.

8. Biological interpretation

- Include functional annotation (eg, target prediction, pathway enrichment) when relevant, and clarify tools/databases used.

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Ethics statement

This review and meta-analysis synthesizes previously published data; therefore, ethics approval and individual patient consent were not required.

Conflict of interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Supplementary materials

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.identj.2025.109305.

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